

What is Machine Learning ?

- A branch of Artificial Intelligence (AI) that enables systems to learn from data.
- Identifies patterns and relationships in training data.
- Uses learned patterns to make predictions on new, unseen data.
- Eliminates the need for explicitly programmed rules.
- Supports automated and intelligent decision-making.

Types of Machine Learning

❑ Supervised Learning

- Uses labeled input-output data.
- Learns the mapping from inputs to known outputs.
- Used for prediction tasks.
- Types:
 - **Classification:** Target label is discrete.
 - **Regression:** Target label is continuous.

❑ Unsupervised Learning

- Uses unlabeled data.
- No predefined output labels are required.